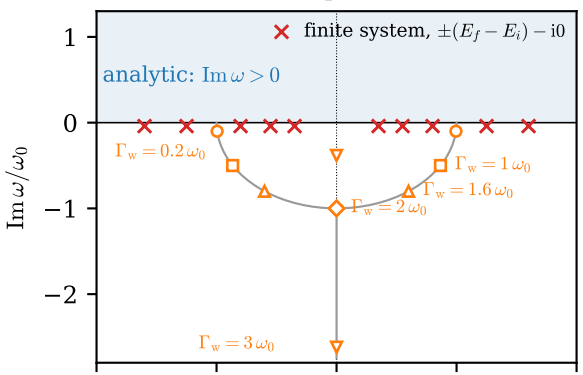
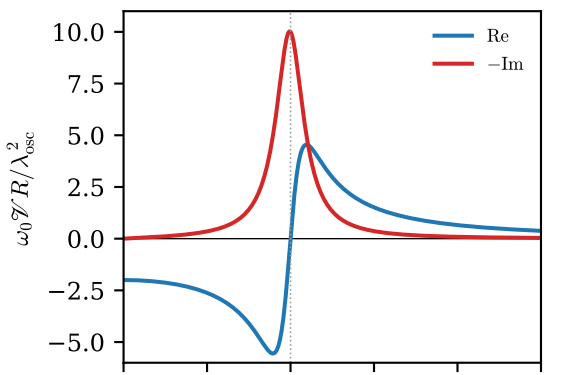

(a) poles
finite system, $\pm(E_f - E_i) - \mathrm{i}0$
analytic: $\mathrm{Im}\,\omega > 0$
$\Gamma_\mathrm{w} = 0.2\,\omega_0$
$\Gamma_\mathrm{w} = 1\,\omega_0$
$\Gamma_\mathrm{w} = 1.6\,\omega_0$
$\Gamma_\mathrm{w} = 2\,\omega_0$
$\Gamma_\mathrm{w} = 3\,\omega_0$
$\mathrm{Im}\,\omega/\omega_0$
$\mathrm{Re}\,\omega/\omega_0$


(b) $\Gamma_\mathrm{w} = 0.2\,\omega_0$
Re
$-$Im
$\omega_0 \mathscr{T} R/\lambda_\mathrm{osc}^2$
ω/ω_0